Hemodynamic Disorders
LG 6, Edema

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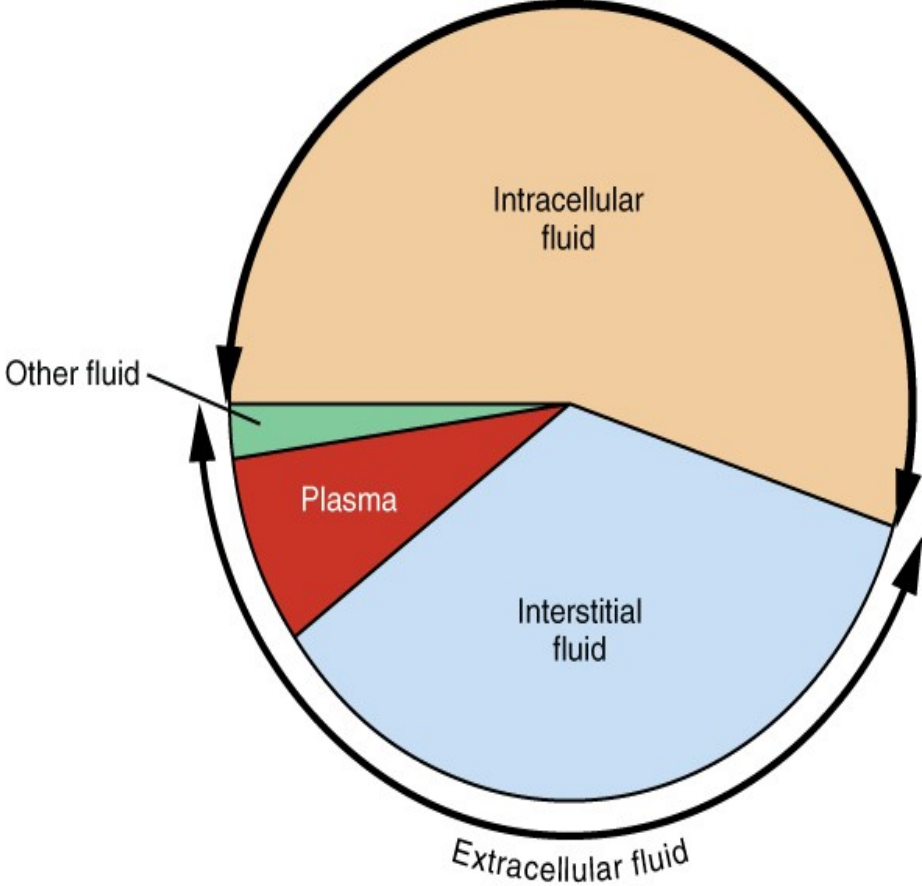
2026-2027

Outline:

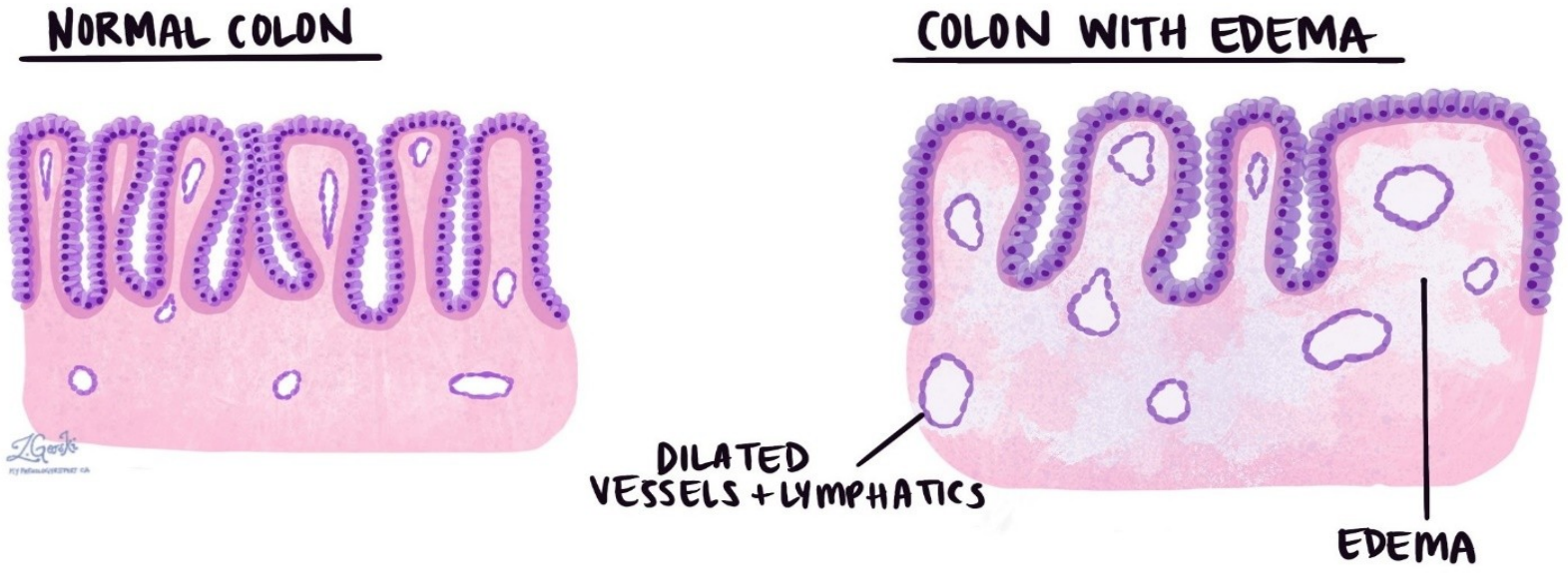
- Introduction
- Definition
- Types of edema fluid
- Causes of Edema
- Morphology
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Introduction

- Approximately 60% of total body weight is water.
- Two thirds of the body's water is intracellular,
- The remainder is in extracellular compartments in the form of interstitial fluid (lies between cells)
- Only about 5% of total body water is in blood plasma.

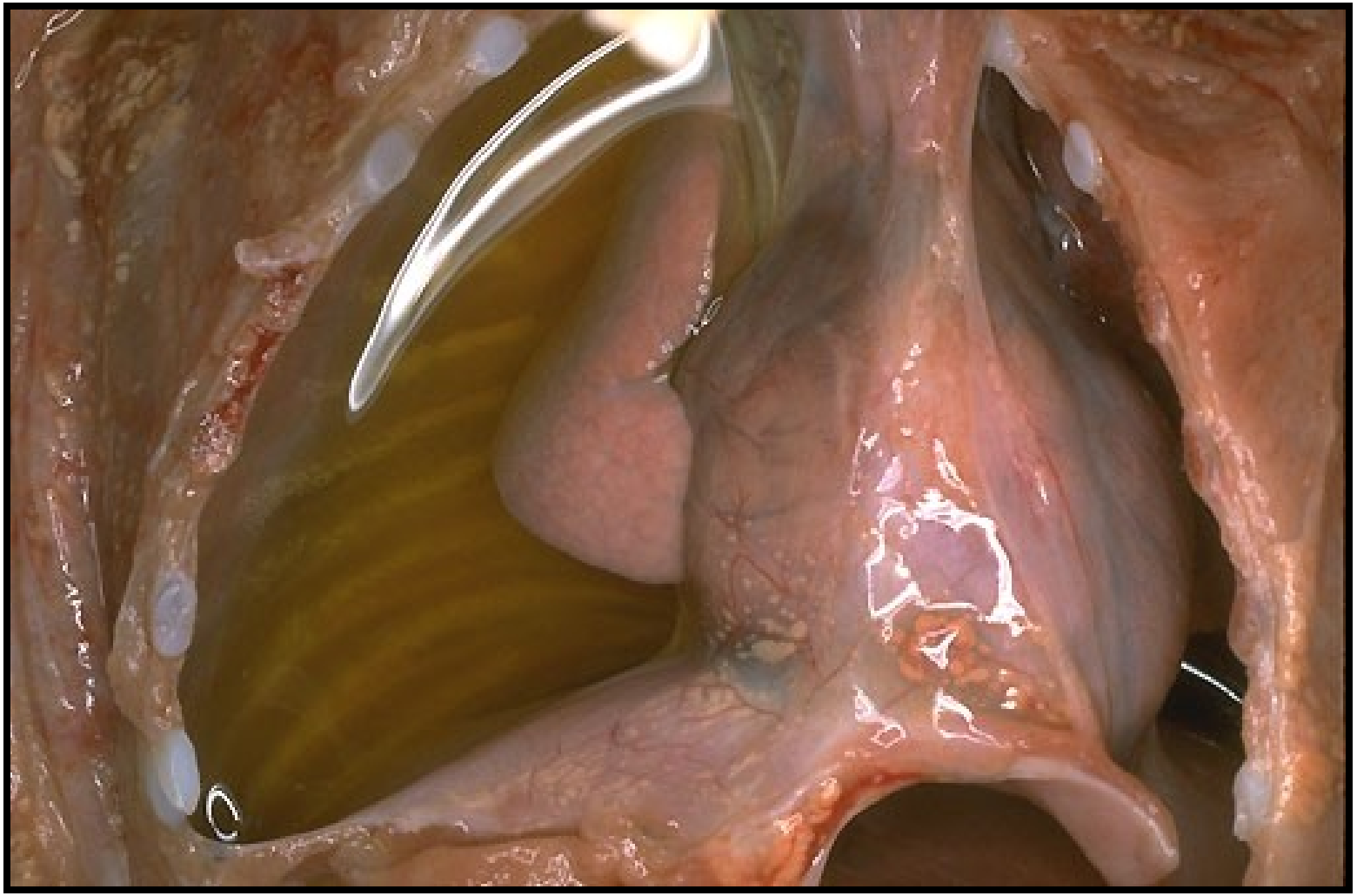


- **edema** is define as an accumulation of interstitial fluid within tissues or in body cavities referred as effusion.



Examples of effusion:

- plueral effusion (Hydrothorax)
- pericardial effusion (hydropericardium))
- ascites (Hydroperitoneum)



Pleural effusion

Ascites



- **Anasarca** is severe, generalized edema marked by profound swelling of subcutaneous tissues and accumulation of fluid in body cavities.



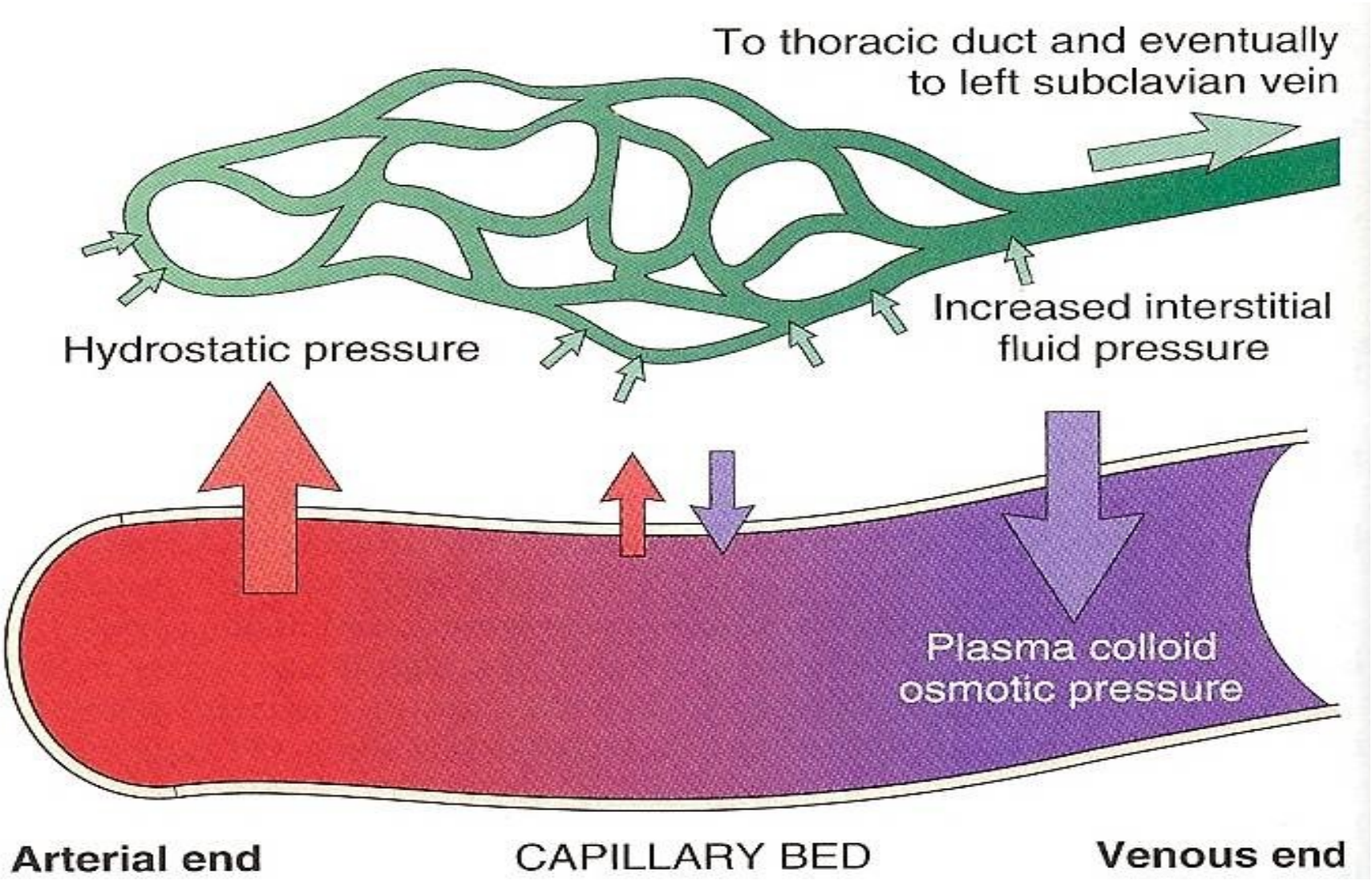
- Fluid movement between the vascular and interstitial spaces is governed mainly by two opposing forces:

1. **the vascular hydrostatic pressure**
2. **the colloid osmotic pressure produced by plasma protein**

- **Normally,** the outflow of fluid produced by hydrostatic pressure at the arteriolar end of the microcirculation is nearly balanced by inflow at the venular end owing to slightly elevated osmotic pressure;

hence there is only a small net outflow of fluid into the interstitial space, which is drained by lymphatic vessels.

- Either increased hydrostatic pressure or diminished colloid osmotic pressure causes increased movement of water into the interstitium.


To thoracic duct and eventually
to left subclavian vein
Increased interstitial
fluid pressure
Hydrostatic pressure
Plasma colloid
osmotic pressure
Arterial end
CAPILLARY BED
Venous end

Types of edema fluid:

1. **Transudate**: fluid that accumulates in the setting of increased hydrostatic pressure or reduced intravascular colloid, it is protein-poor fluid.
2. **Exudate**, (inflammatory edema fluid), it is a protein-rich fluid with a high specific gravity because of increased vascular permeability accompanied inflammation.

Causes of Edema:

1. **Increased hydrostatic pressure**

- **Impaired venous return**
  - ✓ Congestive heart failure
  - ✓ Constrictive pericarditis
  - ✓ Ascites (liver cirrhosis)
  - ✓ Venous obstruction or compression
  - ✓ Thrombosis
  - ✓ External pressure (e.g., mass)
  - ✓ Lower extremity inactivity with prolonged dependency
- **Arteriolar dilation**
  - ✓ Heat
  - ✓ Neurohumoral dysregulation

2. Reduced plasma osmotic pressure (hypoproteinemia)

- ✓ Protein-losing glomerulopathies (nephrotic syndrome)
- ✓ Liver cirrhosis
- ✓ Malnutrition
- ✓ Protein-losing gastroenteropathy

3. Lymphatic obstruction

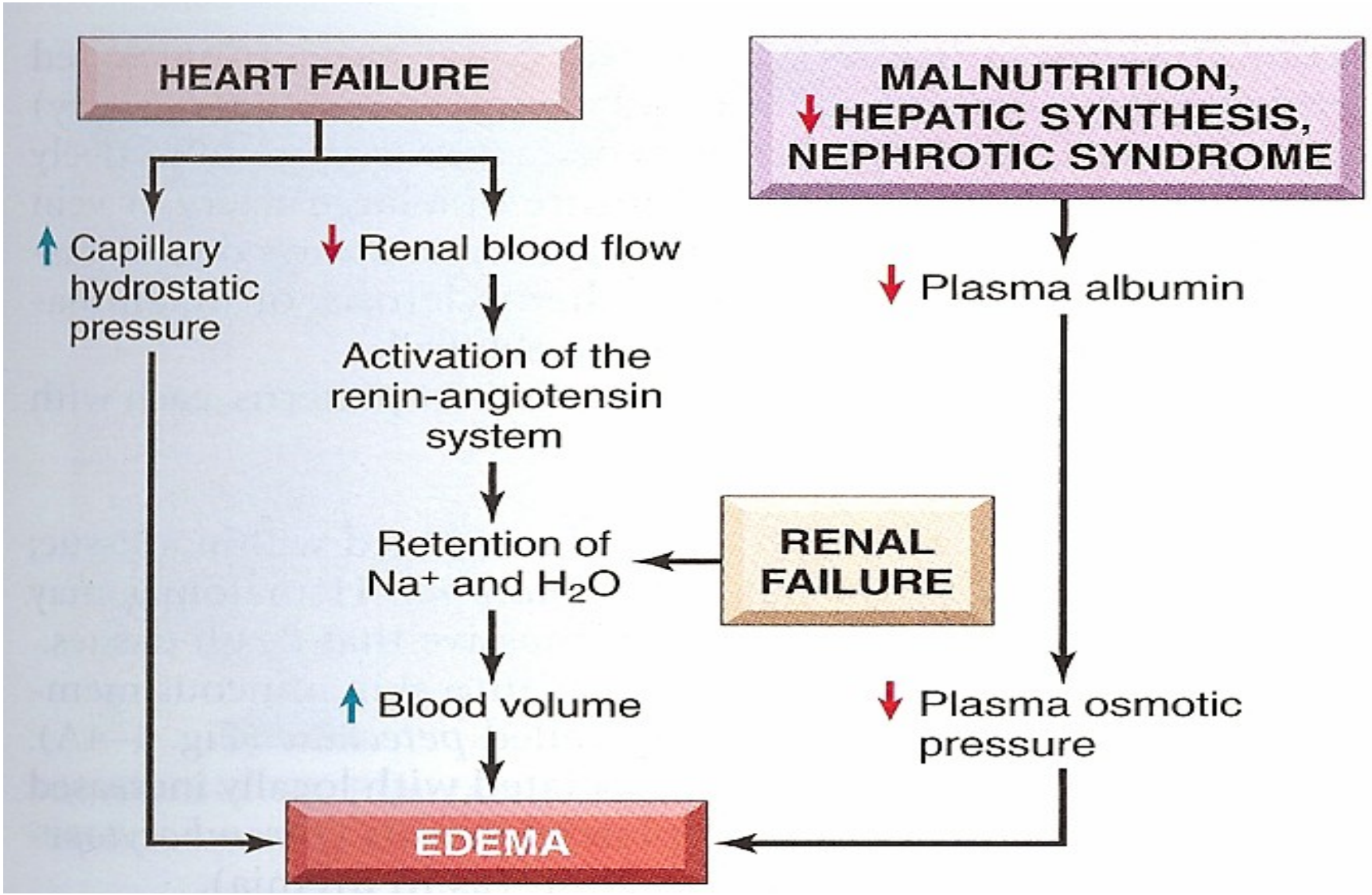
- ✓ Inflammatory
- ✓ Neoplastic
- ✓ Postsurgical
- ✓ Postirradiation
- ✓ Parasitic

4. **Sodium retention**

- Excessive salt intake with renal insufficiency
- Increased tubular reabsorption of sodium
- Renal hypoperfusion
- Increased renin-angiotensin-aldosterone secretion

5. inflammation

- Acute inflammation,
- Chronic inflammation,



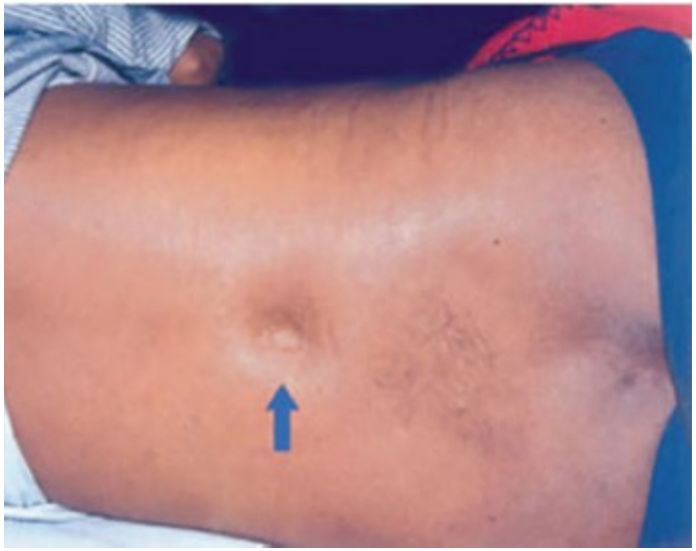
Pathways leading to systemic edema resulting from heart failure, renal failure, or reduced plasma osmotic pressure.

Morphology:


Edema (swelling) of the ankles and feet

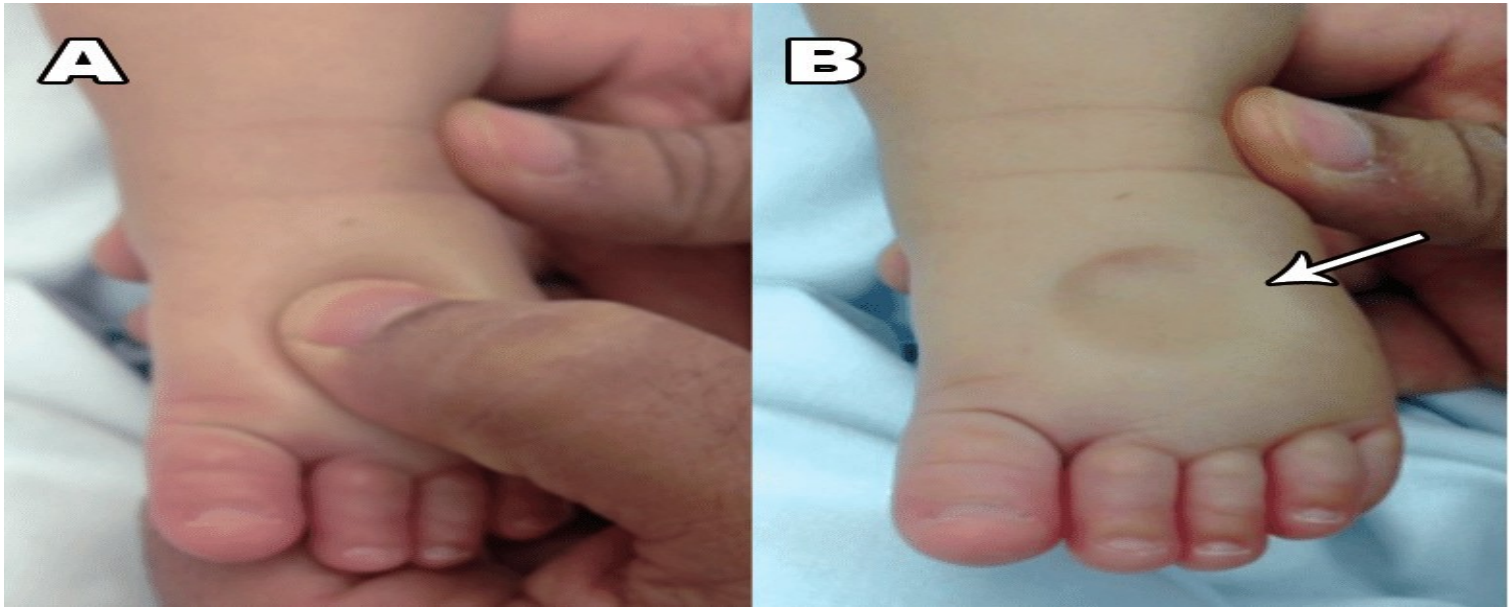
1. **Subcutaneous edema**

- In most cases the distribution is influenced by gravity and is termed dependent edema
- e.g., the legs when standing,
- the sacrum when recumbent)



Pitting edema:

Finger pressure over edematous subcutaneous tissue displaces the interstitial fluid, leaving a finger-shaped depression.



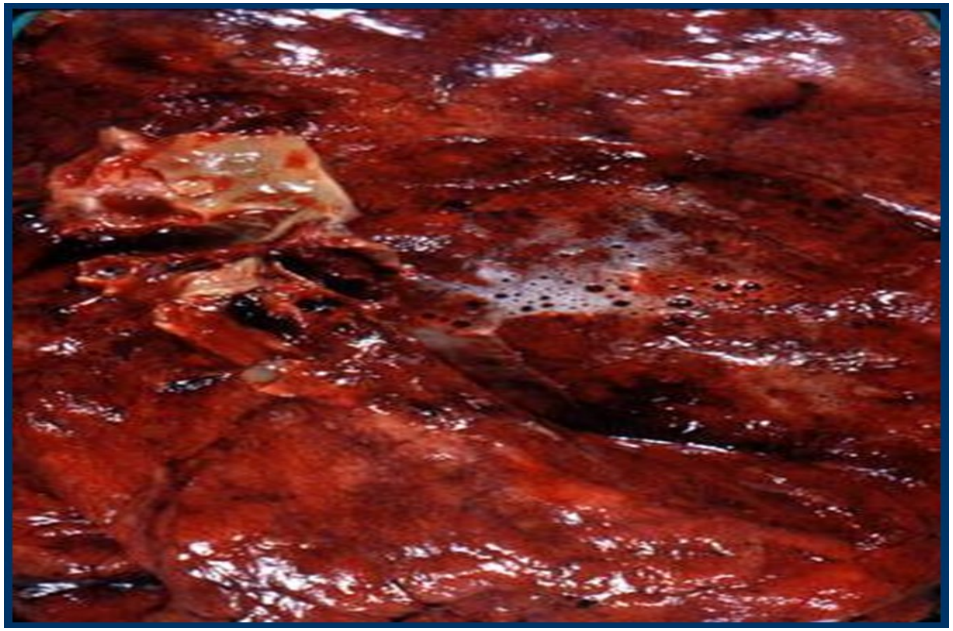
2. **Periorbital edema** is a characteristic finding in severe renal disease.

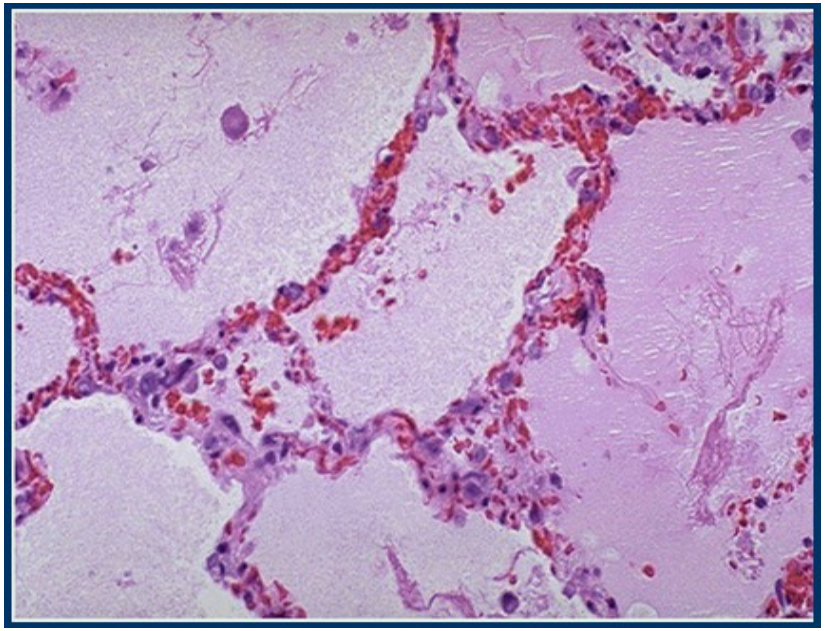
Edema as a result of renal dysfunction can affect all parts of the body BUT It often initially manifests in tissues with loose connective tissue matrix, such as the eyelids.



3. Pulmonary edema

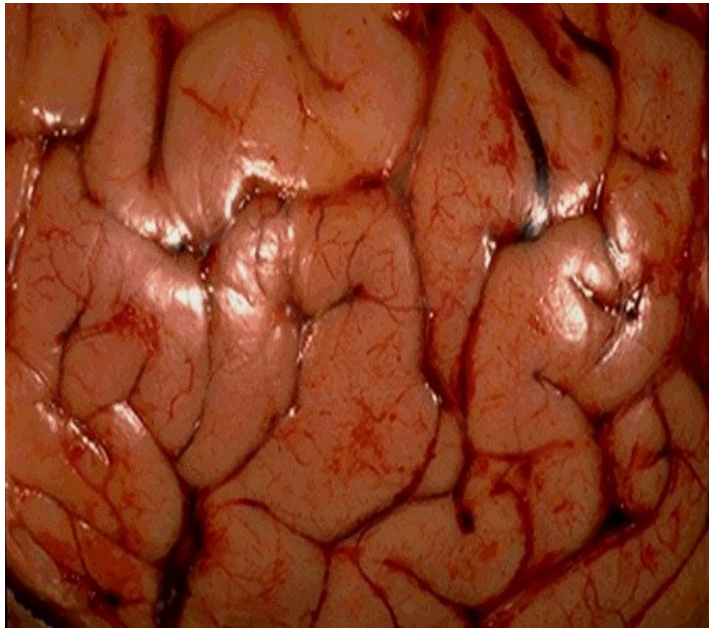
- the lungs are often two to three times their normal weight,
- sectioning yields frothy, blood-tinged fluid—a mixture of air, edema, and extravasated red cells.





4. Brain edema

- can be localized or generalized depending on the nature and extent of the pathologic process or injury.
- With generalized edema the brain is grossly swollen with narrowed sulci; distended gyri show evidence of compression against the unyielding skull.



Clinical Features:

The effects of edema vary, ranging from merely annoying to rapidly fatal:

- ✓ **Subcutaneous edema** is important to recognize primarily because it signals potential underlying cardiac or renal disease; however, when significant, it also can impair wound healing and the clearance of infections.

✓ **Pulmonary edema**

- is a common clinical problem.
- It can cause death by interfering with normal ventilatory function; also creates a favorable environment for infections.

What are causes of pulmonary edema??

✓ **Brain edema**

•is life threatening; if the swelling is severe, the brain can herniate (extrude) through the foramen magnum.

•With increased intracranial pressure, the brain stem vascular supply can be compressed, leading to death due to injury to the medullary centers controlling respiration and other vital function

Summary:

- **Edema** results from the movement of fluid from the vasculature into the interstitial spaces; the fluid may be protein poor (transudate) or protein rich (exudate).
- **Edema may be caused by:**

1. Increased hydrostatic pressure (e.g., heart failure)
2. Increased vascular permeability (e.g., inflammation)
3. Decreased colloid osmotic pressure resulting from reduced plasma albumin
4. Decreased synthesis (e.g., liver disease, protein malnutrition)
5. Increased loss (e.g., nephrotic syndrome)
6. Lymphatic obstruction (e.g., inflammation or neoplasia)
7. Sodium retention (e.g., renal failure)

THANK YOU